

FIT5196 Data Wrangling

Assessment 1 Marking Rubric

This rubric must be read with the A1 student specification and the group's public_data_dictionary.csv. Marks are awarded for demonstration.

1. Performance levels and scoring

Each assessable criterion has its own mark allocation and HD, D, C, P and Fail standard. For a criterion worth M marks, the available criterion-mark ranges are: **HD 80.00-100% of M**; **D 70.00 to below 80.00%**; **C 60.00 to below 70.00%**; **P 50.00 to below 60.00%**; and **Fail 0 to below 50.00%**.

2. Mark allocation summary

Assessment area	Marks
A. Structured parsing and source profiling	1.5
B. Core relational transformation	3.0
C. Multi-source reconciliation and integration	1.5
D. Regex, Unicode and text processing	2.5
E. Validation, reproducibility and efficiency	2.0
F. EDA and finding quality	3.5
G. ML questions and analytical communication	1.0
Total	15.0

3. Performance-standard rubric

A. Structured parsing and source profiling (1.5 marks)

Criterion	Marks	HD	D	C	P	Fail
A1. Structured parsing and source profiling	0.9	Both JSON and XML are parsed with appropriate structured parsers. Source objects, nesting, repeated elements, grains, keys, formats and assumptions are accurately profiled with traceable evidence. No regex is used to reconstruct document structure.	Both sources are parsed correctly and the main structures, grains, keys and formats are accurately profiled. Minor omissions or imprecise assumptions do not affect the transformation design.	Both sources are parsed with a workable method. Core structures and grains are identified, but several profiling details, keys, formats or assumptions are incomplete or contain non-critical errors.	At least one source is parsed sufficiently to support some outputs, but parsing is fragile or profiling is limited. Important structures, grains or assumptions are missing.	One or both sources are not reliably parsed, regex is used as the main structural parser, or there is little credible profiling evidence.
A2. Source-to-target mapping	0.6	95.00-100% of required mapping rows are complete, accurate and traceable across all six required mapping fields. Derivations and overlap rules are specific and consistent with the notebook.	85.00% to below 95.00% of required rows meet the mapping standard. Remaining gaps are minor and do not obscure the overall lineage.	70.00% to below 85.00% of required rows meet the mapping standard. Core lineage is visible, but several paths, derivations, overlap rules or evidence references are incomplete.	50.00% to below 70.00% of required rows meet the mapping standard. The mapping gives partial lineage but is difficult to reproduce or verify.	Below 50.00% of required rows meet the mapping standard, the mapping is missing, or entries largely repeat target names without source lineage.

B. Core relational transformation (3.0 marks)

Criterion	Marks	HD	D	C	P	Fail
B1. Required data products, structure and key coverage	1.0	All six tables are present with the required filenames, grains, fields and field order. The equal-weight mean key-coverage accuracy is 95.00-100%; primary keys are complete and unique.	The equal-weight mean key-coverage accuracy is 90.00% to below 95.00%. All core products are identifiable and structural defects are minor.	The equal-weight mean key-coverage accuracy is 80.00% to below 90.00%. Most products and keys are usable, but there are material omissions, extras or structural inconsistencies.	The equal-weight mean key-coverage accuracy is 70.00% to below 80.00%. Several products or keys are incomplete, but some required relational outputs remain assessable.	Mean key-coverage accuracy is below 70.00%, or the outputs are largely missing, unreadable or not organised at the required grains.
B2. Field-value accuracy	1.2	The equal-weight mean certified field-value accuracy is 95.00-100% under the published normalisation and comparison rules.	Certified field-value accuracy is 90.00% to below 95.00%; errors are limited and localised.	Certified field-value accuracy is 80.00% to below 90.00%; outputs are broadly usable but contain several material transformation or derivation errors.	Certified field-value accuracy is 70.00% to below 80.00%; substantial correction is required, although a meaningful portion of values is correct.	Certified field-value accuracy is below 70.00%, or values cannot be compared because required outputs are absent or unusable.
B3. Types, formulas and relational form	0.8	Dates, timestamps, booleans, numerics, identifiers and prescribed string missing values consistently meet the public contract. GST, discount, delivery and total calculations follow the stated sequence, and one-to-many relationships remain correctly relational.	The contract and formulas are largely correct with minor, localised type, rounding or representation errors. Relational grains remain valid.	Core types and formulas are implemented, but repeated inconsistencies or one material arithmetic/relational error affect part of the outputs.	Some required standardisation and arithmetic is correct, but major type, formula, missing-value or flattening errors remain.	Types and formulas are largely absent or incorrect, or flattening destroys the required relational grains.

C. Multi-source reconciliation and integration (1.5 marks)

Criterion	Marks	HD	D	C	P	Fail
C1. Within-source duplication and cross-source overlap	0.9	Duplicate and overlapping records are detected using stable business keys and field evidence, reconciled reproducibly, and retained once without double counting. Coverage is demonstrated for both within-source and cross-source cases.	The method correctly handles both duplication types with minor coverage or documentation gaps that do not materially affect canonical outputs.	A generally valid reconciliation method is shown, but some overlap cases, fields or source branches are incompletely handled.	Some duplicates are identified, but the method is partial, fragile or reliant on simple concatenation followed by limited cleanup.	Duplication/overlap is ignored, materially double counted, or handled mainly through hard-coded IDs or certified counts.
C2. Conflict handling and relational integrity	0.6	Normalised fields are compared before reconciliation; conflicts and assumptions are recorded rather than silently overwritten. All required primary/foreign-key and one-to-many relationships are valid and reproducible.	Conflict and relationship handling is largely sound, with minor unexplained choices or isolated integrity defects.	Core relationships are retained and a conflict approach is visible, but several assumptions are weakly justified or some integrity defects remain.	A basic precedence or merge rule is used with limited justification; multiple relationships or conflict cases are not reliably handled.	Conflicts are silently or arbitrarily overwritten, or relational integrity is largely invalid or undocumented.

D. Regex, Unicode and text processing (2.5 marks)

Criterion	Marks	HD	D	C	P	Fail
D1. Narrative cleaning and reference extraction	1.2	Cleaning follows the complete published order for entities, markup, wrappers, exhaustive markers, URLs, emoji, whitespace, case and literal NaN. Order, SKU and promotion references are correctly extracted in upper case with valid boundaries and near-match rejection.	The cleaning and extraction pipeline is largely correct; only minor edge cases or one localised step is mishandled.	Core cleaning and all three extraction types are attempted, but several edge cases, ordering rules or boundary conditions are incorrect.	Basic cleaning or extraction works for simple examples, but major required patterns, missing handling or boundary rules are unreliable.	Cleaning/extraction is largely absent, applied before structural parsing, or produces predominantly invalid text and references.
D2. Multilingual and script handling	0.6	Multilingual UTF-8 clean text is preserved; Latin analysis is separately derived from clean text; Latin diacritics are retained; non-Latin letters are accurately detected; literal NaN behaviour is correct.	The multilingual contract is correctly implemented with minor isolated script, punctuation or sentinel errors.	The separate clean and Latin-analysis purposes are understood, but several script, diacritic or sentinel cases are mishandled.	Some multilingual content is preserved, but the analysis field or non-Latin indicator is incomplete or conceptually confused.	Valid multilingual text is erased, non-ASCII is treated as equivalent to non-Latin, or the required derived fields are largely invalid.
D3. Public, student-designed and private functional tests	0.7	All public cases and focused additional tests are demonstrated. Private-test pass rate is 90.00-100%, and submitted functions import and follow the fixed interface.	Most public/additional tests are demonstrated. Private-test pass rate is 75.00% to below 90.00%; failures are limited edge cases.	Public testing is visible but additional coverage is uneven. Private-test pass rate is 50.00% to below 75.00%.	Testing evidence is limited. Private-test pass rate is above 0% and below 50.00%, but some functions operate through the required interface.	Private-test pass rate is 0%, functions are non-importable/interface-incompatible, or there is no credible test evidence.

E. Validation, reproducibility and efficiency (2.0 marks)

Criterion	Marks	HD	D	C	P	Fail
E1. Validation coverage and correctness	1.0	Executable checks correctly cover schema, data types, missing representation, primary/foreign keys, source and row flow, overlap, arithmetic, temporal order, text references, multilingual preservation and relevant ranges. Tolerances and observed results are explicit.	Nearly all required validation areas are correctly covered; omissions or weak checks are minor and do not conceal a material defect.	Most core validation areas are covered, but several checks are shallow, incomplete or use weak tolerances.	A basic validation set runs, but major required areas are absent or checks mostly print summaries without testing behaviour.	Validation is absent, largely non-executable, or relies mainly on hard-coded certified counts/answers.
E2. Interpretation, fresh rerun, efficiency and dependencies	1.0	The validation register uses stable IDs, observed results, PASS/FAIL, evidence and justified resolution. Both notebooks complete Restart and Run All offline from configurable paths; outputs are recreated without manual edits. Methods are appropriate to scale and dependencies are declared.	Interpretation and both reruns are largely complete, with minor path, dependency, output or efficiency issues requiring little intervention.	The main workflows are reproducible, but one notebook or several checks require limited repair; interpretation or efficiency is uneven.	Substantial marker intervention is required, but meaningful code, outputs and validation evidence can still be followed.	Workflows cannot be reproduced, depend on unavailable live/private resources, or provide little usable interpretation or executable evidence.

F. EDA and finding quality (3.5 marks)

Criterion	Marks	HD	D	C	P	Fail
F1. Analytical coverage, grain, joins and metrics	1.3	6-8 assessed visualisations cover all six required categories, at least four tables and at least two correct relational analyses. Questions, observation units, denominators, aggregations and join keys are correct, with explicit checks against row multiplication.	6-8 visualisations cover at least five categories, at least four tables and two relational analyses. Grain and metrics are largely correct with minor local errors.	5-8 visualisations cover at least four categories and three tables, including at least one valid relational analysis. Several grain, denominator or join explanations need improvement.	At least four visualisations cover at least three categories and two tables. Analytical questions are recognisable, but joins, grains or denominators contain major weaknesses.	Fewer than four assessable visualisations, predominantly invalid analysis, or little evidence that reported metrics can be reproduced.
F2. Visual design and readability	0.6	Chart forms fit the questions and data; labels, units, scales, legends and coordinated panels are accurate, readable at 100% zoom and free from materially misleading design choices.	Visuals are clear and appropriate with only minor labelling, scale, density or presentation issues.	Most visuals are interpretable, but several chart choices, labels, scales or layouts reduce analytical clarity.	Some visuals communicate basic results, but major readability or design problems make interpretation difficult.	Visuals are absent, largely unreadable, materially misleading or disconnected from the stated questions.

Criterion	Marks	HD	D	C	P	Fail
F3. Evidence-based findings, interpretation and decision value	1.6	9-10 of the first ten findings are strongly supported by a figure/statistic and correctly state grain, magnitude/denominator, context, uncertainty or alternative explanation, and a proportionate implication. Association is not presented as causation.	At least eight findings are well supported and decision-relevant; most include magnitude and uncertainty, with only minor omissions or overstatement.	6-7 findings are adequately supported, or a larger set is uneven. Business meaning is present, but quantified evidence, uncertainty or implications are sometimes weak.	Five findings have identifiable evidence and basic interpretation, but support, uncertainty and decision value are limited.	Fewer than five findings are supportable, or findings are predominantly unquantified, incorrect, causal without basis or disconnected from the EDA.

G. ML questions and analytical communication (1.0 mark)

Criterion	Marks	HD	D	C	P	Fail
G1. ML question framing, evaluation, risk and communication	1.0	All five questions are feasible, grounded in EDA and collectively cover at least two problem types. Each specifies the decision, unit, target/objective, decision-time predictors, validation split, metric and relevant leakage/fairness/deployment risks. Communication is concise and external material is consistently referenced.	Five feasible questions are provided and most required design/risk elements are correct; omissions are minor and communication is clear.	Four usable questions are provided, or five are uneven. Core targets, predictors and evaluation ideas are present, but several validation or risk elements are underdeveloped.	Three basic, relevant questions are identifiable, but framing, evaluation design, decision-time availability or risk analysis is limited.	Fewer than three usable questions, questions are infeasible or not grounded in the available data, or major leakage/evaluation problems are unrecognised.

4. Quantitative scoring rules

4.1 Mapping completeness

For A2, the denominator is the number of required pre-filled target rows in the mapping template. A row counts as meeting the mapping standard only when the applicable entries across source_format, JSON/XML paths, transformation_or_derivation, overlap_or_conflict_rule and notebook_evidence are accurate and sufficient for that target field. N/A is acceptable only when justified by the source and target contract.

4.2 Key coverage and field-value accuracy

B1 and B2 are calculated separately for each of the six required tables, then averaged using equal table weights so a large table does not dominate the mark. The performance thresholds are:

Accuracy	Performance level
95.00-100%	HD
90.00% to below 95.00%	D
80.00% to below 90.00%	C
70.00% to below 80.00%	P
Below 70.00%	Fail

A missing or unreadable table scores zero for that table's B1 and B2 shares; other tables remain eligible for credit. Structural conformance, key coverage and field values are assessed separately. Comparisons use the published keys, normalisation rules, field order, null handling and numeric tolerances.

4.3 Private functional tests

The private-test component of D3 is worth 0.4 of the 0.7 criterion marks:

Private-test pass rate	Private-test component
90.00-100%	0.4
75.00% to below 90.00%	0.3
50.00% to below 75.00%	0.2
Above 0% to below 50.00%	0.1
0% or non-importable/interface-incompatible functions	0.0

Private tests only assess behaviour published in the specification and public test contract. The remaining 0.3 D3 marks assess demonstrated public cases and student-designed tests.

4.4 Count-limited evidence

Only the first eight clearly labelled assessed visualisations, first ten numbered findings and first five numbered ML questions are marked. Additional items do not replace an earlier weak item and do not create bonus marks.

5. Evidence, partial credit and fairness

- A missing or misnamed artefact loses only the credit that cannot be unambiguously evidenced from the remaining submission.
- A correct value without reproducible code does not receive full method or reproducibility credit.
- A defensible method with a local implementation error can receive method credit within the applicable descriptor.
- EDA based on an imperfect student output can receive reasoning credit when its grain, calculations and interpretation are valid for that output. Claims whose validity depends on the error cannot receive full accuracy credit.
- A genuine failed validation check can receive validation and interpretation credit when it correctly identifies the issue and proposes a justified treatment. Fabricating a value to make a check pass receives no such credit.
- The same technical defect is not repeatedly penalised as several independent errors. Distinct downstream effects may be considered where they assess a different learning outcome.
- AI use is permitted as stated in the specification but is not a marking criterion. Missing declarations or integrity concerns are handled through the applicable University process, not an unpublished rubric deduction.

6. Criterion-to-task alignment

Specification task	Rubric criteria
Task 1: Parse and profile	A1, A2, E1-E2
Task 2: Relational outputs	B1-B3, C1-C2, E1-E2
Task 3: Regex and multilingual text	D1-D3, E1-E2
Task 4: Validation	C2, E1-E2
Task 5: EDA	F1-F3
Task 6: Findings and ML questions	F3, G1